

UNIFIED MILITARY ANALYTICS AND COMPARISON DASHBOARD – DV

Infosys Springboard Virtual Internship 6.0

Comprehensive Technical Documentation Report

This document presents complete milestone-wise technical implementation details, architecture decisions, KPI logic, validation strategy, and final delivery outcomes.

EXECUTIVE SUMMARY:

The Unified Military Analytics and Comparison Dashboard (DV) is an end-to-end analytics platform designed to collect, standardize, analyze, and visualize global military power data for 2025. The solution integrates automated web scraping, data engineering pipelines, KPI computation, interactive dashboard development, validation testing, and repository packaging into a unified workflow.

The system processes 140+ countries and 50+ defense and economic indicators, delivering four integrated dashboards: Quick Stats, Nation Overview, Compare Powers, and Coalition Builder.

SYSTEM ARCHITECTURE OVERVIEW

The project follows a structured analytics pipeline:

1. Data Acquisition (Web Scraping)
2. Data Cleaning & Standardization
3. KPI Engineering & Metadata Enrichment
4. Dashboard Development & Integration
5. Testing & Validation
6. Documentation & Deployment

1. MILESTONE 1 – DATA COLLECTION AND PREPARATION:

1.1 Module 1 – Scraping Setup and Execution

Objective: Automate extraction of structured military metrics.

Implementation Details:

- Python libraries used: requests, BeautifulSoup
- Input source: predefined URL list
- Parsing logic extracted country names and metric values
- Structured dictionary converted to pandas DataFrame
- Output stored as military_raw_data.csv

Validation Criteria Achieved:

- ≥ 95% successful URL parsing
- 140+ country records generated

Deliverables:

- Script: scrape_military_metrics.py
- Output: military_raw_data.csv

1.2 Module 2 – Data Cleaning and Structuring

Objective: Convert raw text-based metrics into analysis-ready dataset.

Cleaning Steps:

- Removed commas, %, +, special symbols
- Converted text fields to numeric values
- Standardized column naming conventions
- Handled missing values
- Verified numeric consistency

Final Output: military_cleaned.csv

Missing Data Rate: < 2%

Deliverables:

- military_cleaned.csv
- Python notebook: clean_data.ipynb

2. MILESTONE 2 – KPI ENGINEERING & DASHBOARD PREPARATION

2.1 Module 3 – KPI Feature Engineering

Objective: Transform cleaned data into analytical KPIs.

KPIs Engineered:

- Assets per Capita = Total Military Assets / Population
- Budget per Soldier = Defense Budget / Total Military Manpower
- Budget-to-GDP Ratio
- Power Index Rank Gap
- Domain Contribution Indices (Air, Land, Naval, Economic)
- Military Efficiency Index

Normalization was applied to ensure fair comparison across countries. Dense ranking was used to compute overall rankings.

Output: military_final.xlsx (Wide & Long format)

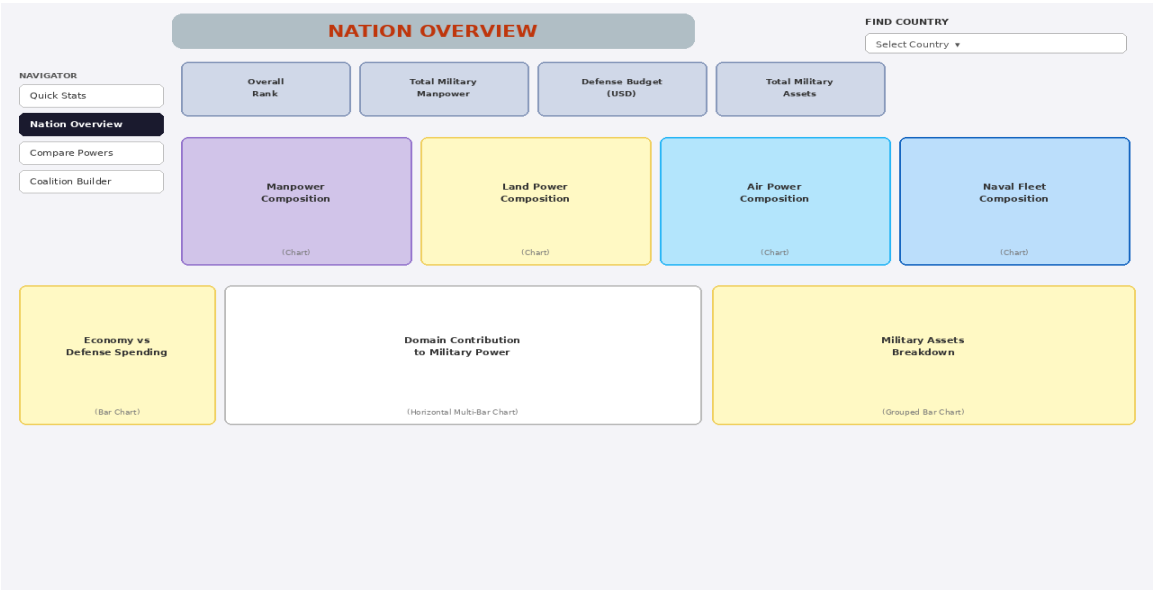
2.2 Module 4 – Dashboard Planning & Prototyping

Wireframes defined layout structure, KPI cards, slicers, navigation logic, and cross-filter interactions. Prototype dashboard validated KPI mapping.

-Quick Stats Draft Dashboard:



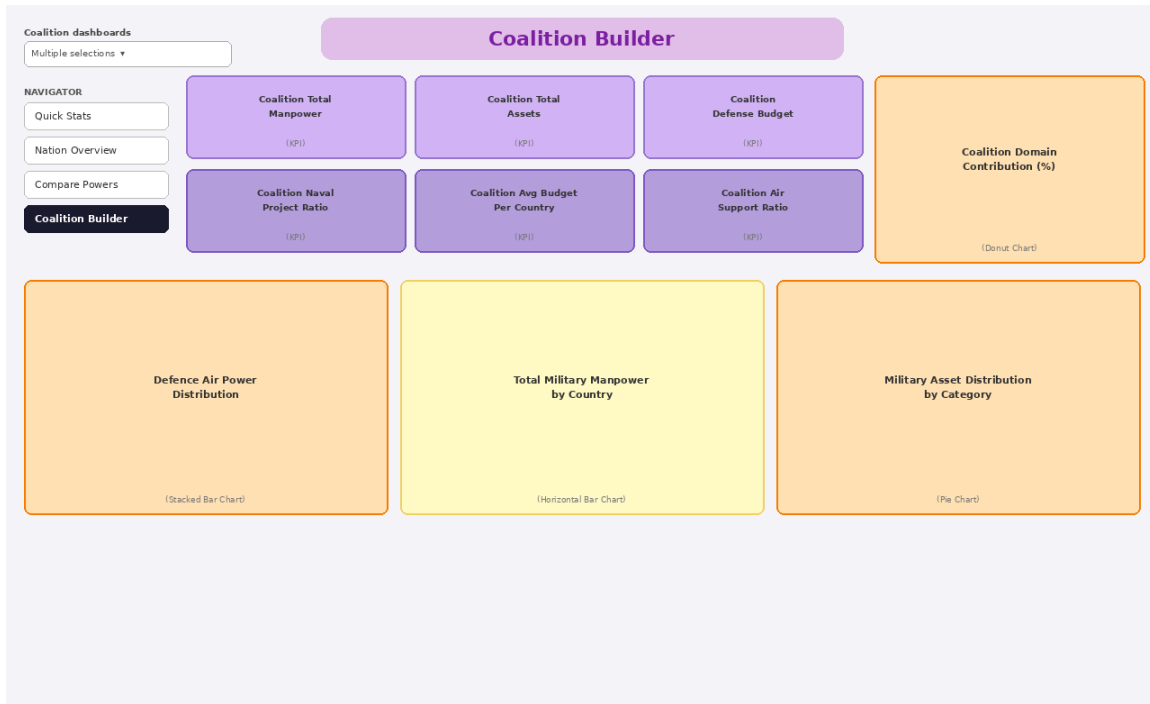
-Nation Overview Draft Dashboard:



- Compare Powers Draft Dashboard:



Coalition Builder Draft Dashboard:



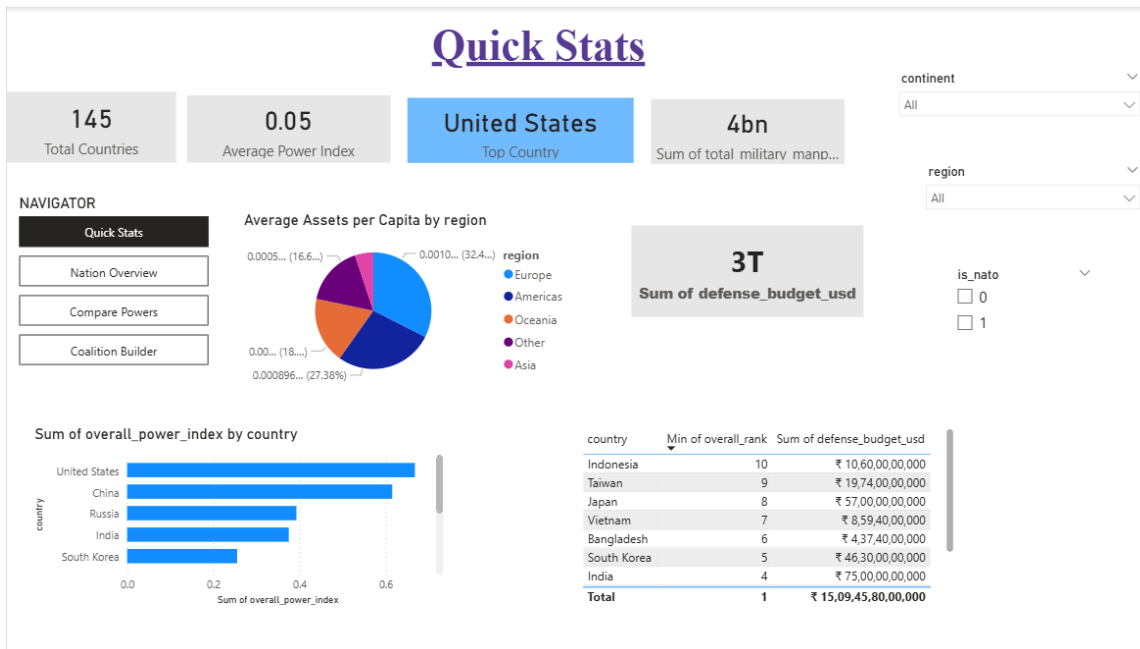
3. MILESTONE 3 – FULL DASHBOARD DEVELOPMENT

3.1 Quick Stats Dashboard

Purpose: Provide global summary of military power distribution.

Core Elements:

- Total Countries KPI
- Average Power Index
- Top Ranked Country
- Total Defense Budget
- Top 10 Overall Power Chart
- Region & NATO Filters

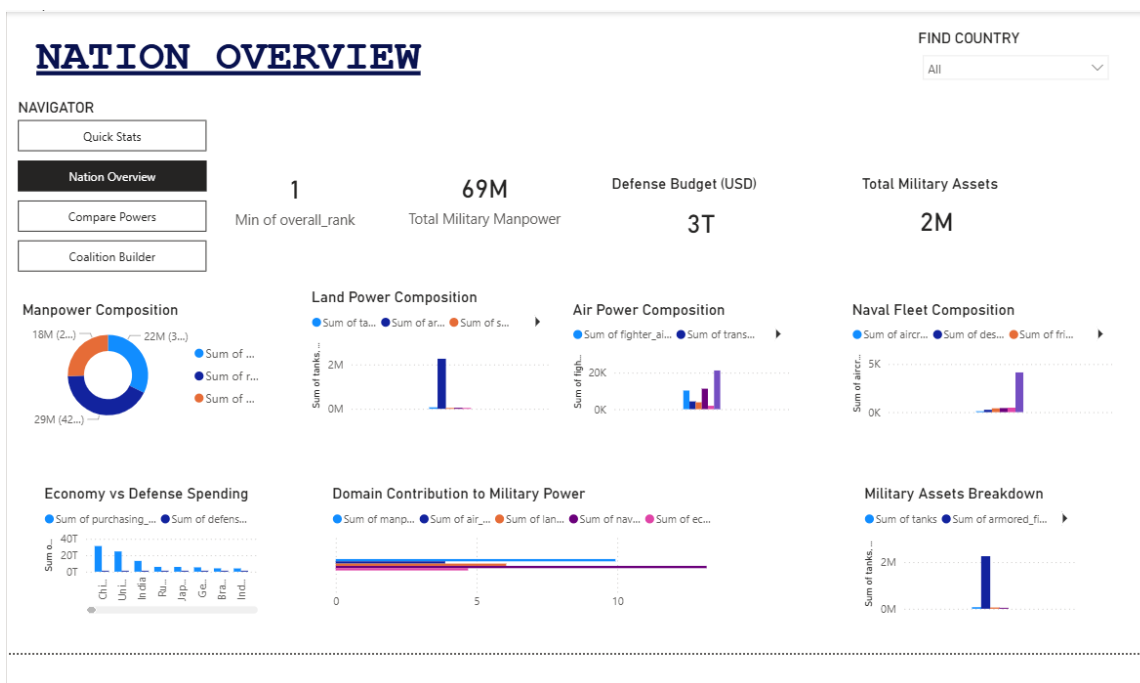


3.2 Nation Overview Dashboard

Purpose: Single-country detailed breakdown.

Components:

- Overall Rank
- Manpower Composition
- Land, Air, Naval Asset Charts
- Economy vs Defense Spending
- Domain Contribution Visualization

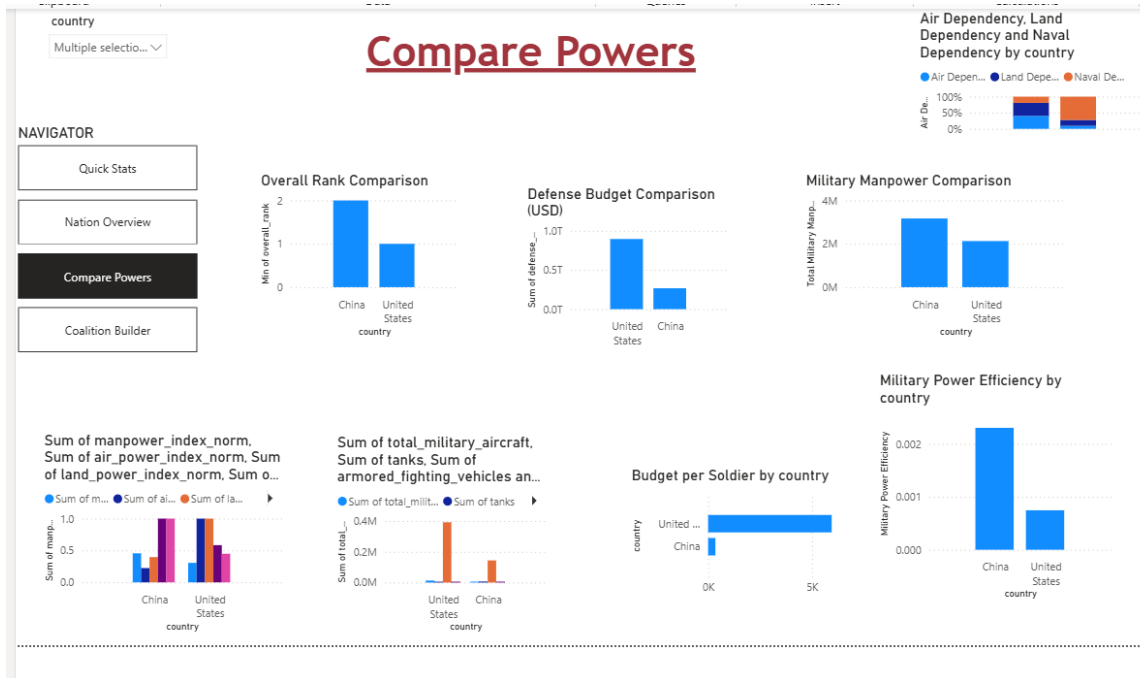


3.3 Compare Powers Dashboard:

Purpose: Two-country comparison.

Comparison Metrics:

- Rank Comparison
- Defense Budget Comparison
- Manpower Comparison
- Efficiency Comparison
- Domain Dependency Ratios

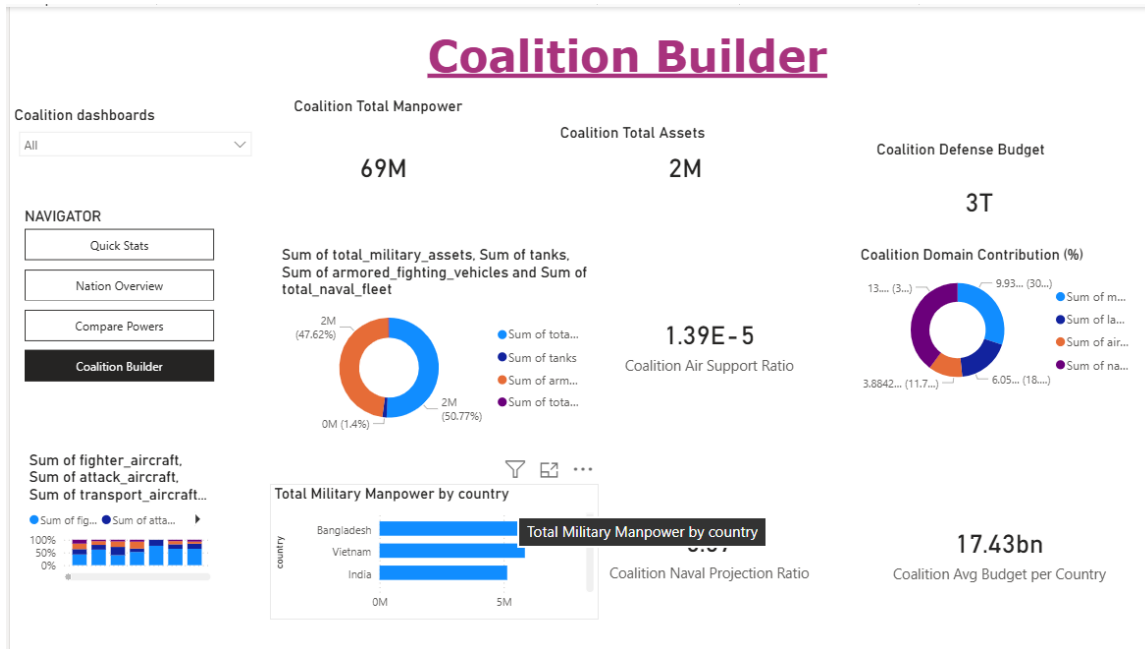


3.4 Coalition Builder Dashboard

Purpose: Multi-country coalition simulation.

Coalition KPIs:

- Coalition Total Manpower
- Coalition Total Assets
- Coalition Defense Budget
- Air Support Ratio
- Naval Projection Ratio
- Average Budget per Country



4. MILESTONE 4 – FINAL REVIEW AND DELIVERY

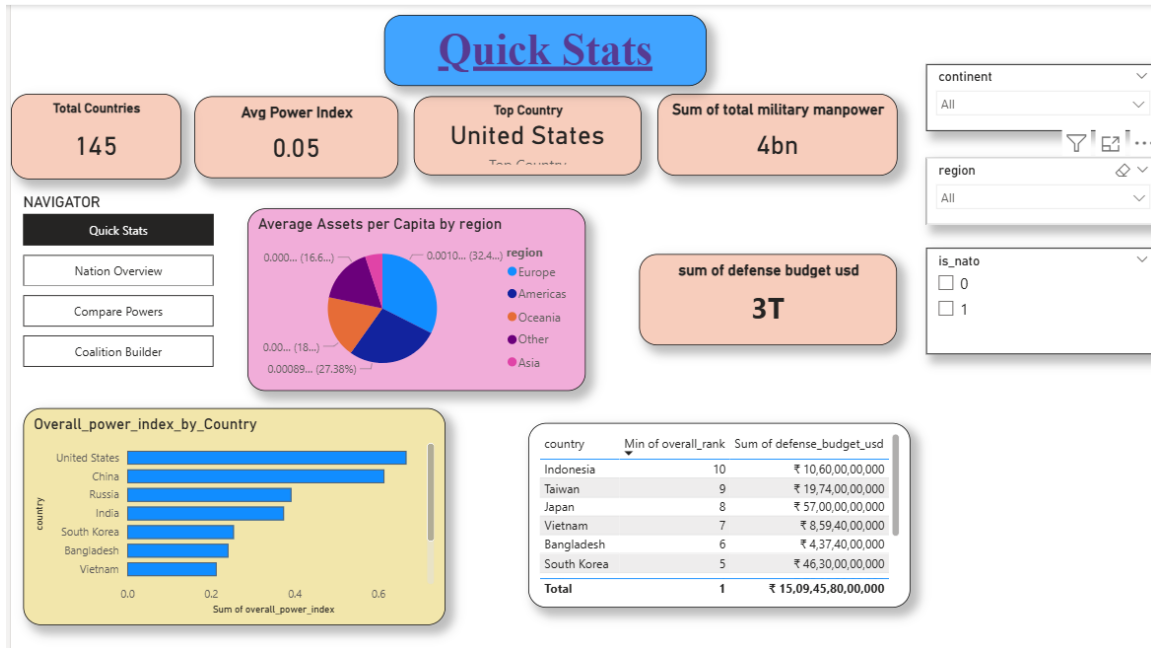
4.1 Testing & Validation Strategy

Validation Process:

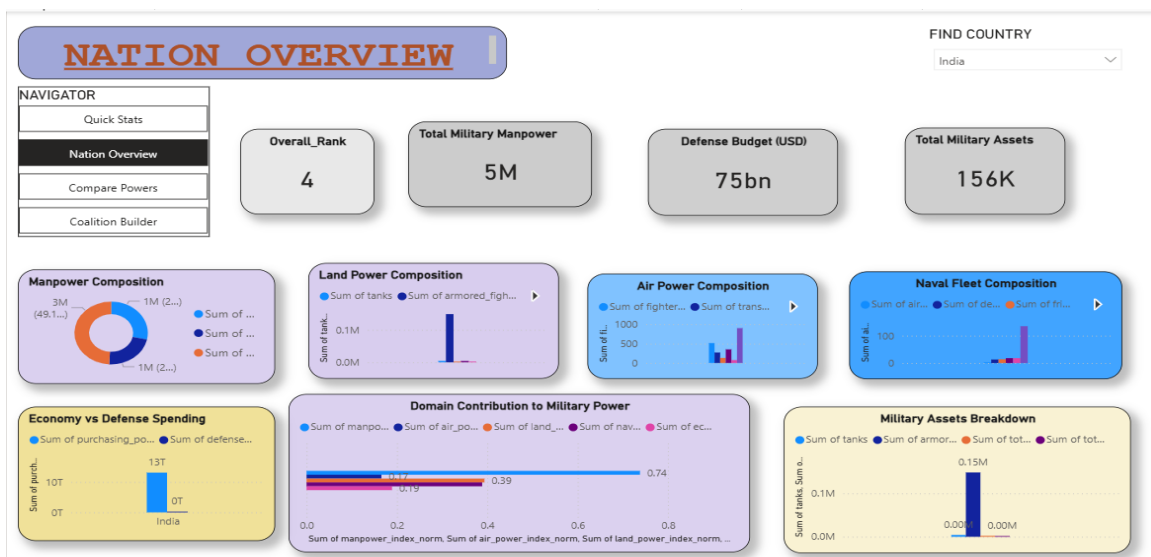
- Tested slicers and filters across pages
- Verified DAX measures
- Manual aggregation cross-checks
- Checked division-by-zero cases
- Verified navigation buttons
- Corrected tooltip and labeling issues

Result: No functional bugs detected.

Quick Stats Final Dashboard:



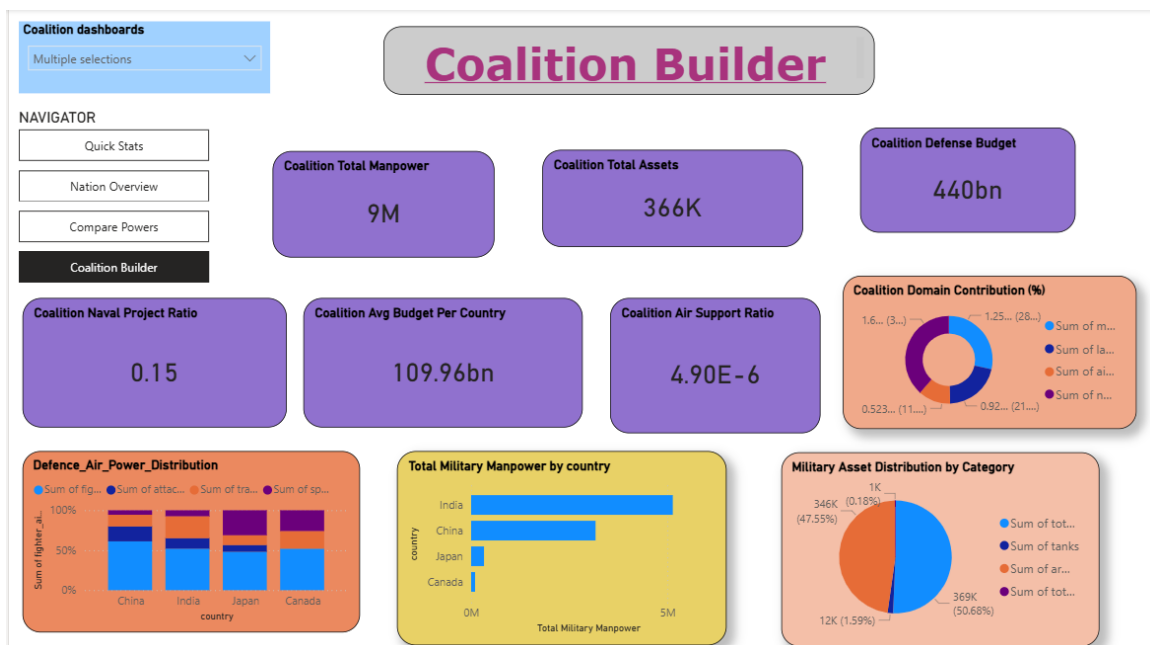
Nation Overview Final Dashboard:



Compare Powers Final Dashboard:



Coalition Builder Final Dashboard:



5.2 Documentation & Repository Packaging

Repository organized into milestone folders:

- Milestone 1: Scraping & Cleaning
- Milestone 2: KPI Engineering
- Milestone 3: Dashboard Development
- Milestone 4: Testing & Documentation

6. TECH STACK

Scraping: Python, requests, BeautifulSoup

Processing: pandas, numpy

Visualization: Microsoft Power BI Desktop

Version Control: GitHub

Documentation: Markdown, MS Word

8. CONCLUSION

The Unified Military Analytics and Comparison Dashboard successfully demonstrates a complete end-to-end analytics implementation lifecycle. The project encompasses automated data acquisition, structured data engineering, KPI development, interactive dashboard integration, systematic validation, and organized repository deployment.

All milestone objectives defined in the internship guidelines have been achieved, including accurate data extraction, standardized dataset preparation, reliable KPI computation, functional dashboard integration, and comprehensive quality assurance testing.

The final deliverable is a fully integrated, validated, and documented analytics solution that is suitable for academic evaluation, professional presentation, and portfolio demonstration.